Constancy RoPE: An Empirical Study of Spectral Attention Gains at 4K

Working research report

October 17, 2026

Abstract

We test whether spectral feedback on rotary-position attention gains improves training-free 4K generation with FLUX.1-Krea-dev. Constancy RoPE compares the predicted clean latent spectrum with a native-resolution reference and applies a bounded, noise-gated correction to spatial attention gains. Two completed campaigns fail to demonstrate an improvement over the released SEGA baseline. In the second campaign, only SEGA plus time-matched constancy at gain $\eta = 1$ passes the eight-prompt development gate. On 187 held-out prompt pairs, its ImageReward difference is $+0.00857$ with a 95% confidence interval of $[-0.00841, 0.02742]$, while full-resolution MUSIQ changes by -0.20860 $[-0.70887, 0.31512]$. PickScore and crop MUSIQ decrease. These results do not support an efficacy or state-of-the-art claim. This working report records the completed experiments; it is not a submission-ready methods paper.

1 Question and tested controllers

The project asks whether feedback toward a model's native latent spectrum can improve resolution extrapolation through per-band attention gains. The target image size is 4096×4096, compared with a 1024×1024 native calibration size. The baseline is the released SEGA [12] allocation and RoPE base, identified in the implementation as `sega_official`. The released allocation differs from the paper-equation implementation; the two are not interchangeable baselines.

The first campaign tested the original constancy controller, then a time-matched native spectral target. The latter was also tested on SEGA's RoPE base with its reference temperature. This did not establish an improvement. The second campaign tested two effective-pool temperature rules, `alias_temp` and `alias_temp_sref`, and a composition of released SEGA with time-matched constancy, `sega_ctm`, at $\eta = 0.5$ and 1.0.

For the composed controller, let m_d be SEGA's released query/key multiplier for rotary pair d. A correction in spectral band b is

$$e_b = \tfrac{1}{2}\big(\log P_{\mathrm{native},b}(\sigma) + a - \log \widehat{P}_b\big), \tag{1}$$

$$\delta_b = \eta w_b \ \mathrm{clip}(e_b, -1, 1), \tag{2}$$

$$g_d = m_d^2 \exp\big(\mathrm{interp}(\delta)_d\big). \tag{3}$$

Here a is the anchor offset, w_b is the existing visibility gate, and $\widehat{P}_b$ is measured from the preceding transformer call's clean estimate. The native target is interpolated at the current scheduler noise level. Query and key are each multiplied by $m_d \exp(\delta_d/2)$. The first step has no correction, the index axis retains unit gain, and no additional global temperature is applied. Spectral powers use orthonormal FFTs and means within bands. The vision analogy motivates this controller; it does not establish its efficacy or novelty.

2 What the gain algebra establishes

Pair scaling and its causal limit. For fixed content vectors, the rotary-position logit [13] can be written as a finite trigonometric sum

$$\ell(\Delta) = \frac{1}{\sqrt{d_k}} \sum_j \mathrm{Re}\big[c_j e^{i\theta_j \Delta}\big]. \tag{4}$$

Applying a shared multiplier m_j to the query and key coordinates of pair j multiplies that term by m_j^2. Its sign and phase still depend on c_j. Increasing a high-frequency coefficient therefore does not by itself establish more local attention, higher image-band power, or sharper decoded detail. The three existing Krea development probes have inconsistent response signs: one prompt has no positive decoded-width diagonal among six tested bands. Those probes jointly perturb both spatial axes on NTK2, so they also cannot identify a single-axis response under released SEGA.

Standardized profile shape. Let ℓ be a nonconstant vector of sampled log powers and let $z(\ell) = (\ell - \operatorname{mean} \ell)/\operatorname{std}(\ell)$, using one fixed standard-deviation convention. For $\beta > 0$,

$$z(a\mathbf{1} + \beta\ell) = z(\ell). \tag{5}$$

This elementary identity applies to an ideal standardized-profile allocation. It does not imply invariance of SEGA's radial spread or its complete gain. The paper-inspired reference implementation adds a standard-deviation epsilon, and the released implementation floors energy and standard deviation. Their behavior is therefore invariant only under the corresponding inactive-floor or negligible-stabilizer conditions, up to floating-point error. The statement is about a formed log-power profile; raising a two-dimensional spectrum to a power need not commute with averaging it into that profile.

Allocation and scalar magnitude are coupled. Before clipping, write a centered SEGA multiplier as $m_j = m_{\mathrm{ref}}(1 + a_j)$ with $\operatorname{mean} a_j = 0$. Then

$$\operatorname{mean}(m_j^2) = m_{\mathrm{ref}}^2\big[1 + \operatorname{mean}(a_j^2)\big]. \tag{6}$$

Thus centering query/key multipliers does not preserve their mean squared magnitude, which is the mean coefficient gain. Clipping changes these moments again. The new RMS controls isolate this effect per spatial axis. They do not match the actual attention-logit distribution, its variance, or its entropy.

Forward-noise expectation and the gate. For the linear data–noise interpolation used to motivate rectified flow [8], assume independent data x_0 and unit white noise ε, with $x_\sigma = (1 - \sigma)x_0 + \sigma\varepsilon$. With an orthonormal transform, channel summation over C channels, and mean-per-band density on non-DC bands,

$$\mathbb{E}_\varepsilon P_b(x_\sigma) = (1 - \sigma)^2 P_b(x_0) + \sigma^2 C. \tag{7}$$

The cross term vanishes in expectation. This is not an exact decomposition of one noisy sample or one generated ODE trajectory. The controller's statistic $\rho_b = \max(P_b(x_\sigma)/(\sigma^2 C) - 1, 0)$ and gate $w_b = \rho_b/(1 + \rho_b)$ are design choices inspired by this identity. They do not estimate the uncertainty of the delayed clean prediction. Likewise, zero correction at an anchored target spectrum is an algebraic property of the rule, not a stability theorem for the full sampling process.

3 Executed protocol

All reported comparisons use the FLUX.1-Krea-dev transformer with FLUX.1-dev text encoders and VAE, 4096^2 output, 28 sampling steps, distilled guidance 4.5, and scheduler shift $\mu = 1.15$. The selected candidate and baseline share the executed schedule, seed convention, and SEGA RoPE base. Earlier FLUX.1-dev outputs exhibited a lattice artifact and are excluded from these results.

The 195 Aesthetic-4K evaluation captions are split into development rows 0–7 and held-out rows 8–194. Native calibration uses 16 separate Pick-a-Pic prompts. Development eligibility requires positive mean differences in both ImageReward and full-resolution MUSIQ, with CLIP difference at least -0.3. At most two variants may advance; no settings change after selection.

The second campaign reports exactly 187 matched seed-0 prompt pairs. Generation produced 188 images because row 7 was included to preserve task chunking; that row is excluded from all held-out tables. Confidence intervals use 2,000 paired bootstrap draws with bootstrap seed 0 and the 2.5 and 97.5 percentiles. Intervals are nominal per metric, without multiple-comparison adjustment. An interval crossing zero does not establish equivalence.

4 Results

4.1 Development selection

Table 1 records selection only. Both alias variants lose ImageReward. The $\eta = 0.5$ composition misses the MUSIQ gate by 0.007850. Only $\eta = 1$ advances; development scores are not generalization evidence.

Table 1: Development means on eight prompts, seed 0.

Variant	ImageReward	MUSIQ	CLIP	Advance
Released SEGA	1.423018	42.488060	30.320004	Reference
Alias temperature	1.281098	35.807356	29.366986	No
Alias, reference gain	1.277426	44.972845	30.238460	No
SEGA + constancy, $\eta = 0.5$	1.441334	42.480210	30.200250	No
SEGA + constancy, $\eta = 1$	1.448742	42.505928	30.342451	Yes

4.2 Held-out comparison

Table 2: Held-out seed-0 means and paired differences, candidate minus released SEGA, over 187 prompts. Higher is better except the last three rows.

Metric	SEGA	SEGA + constancy	Difference [95% CI]
ImageReward	1.18994	1.19852	$+0.00857$ $[-0.00841, +0.02742]$
PickScore	23.05255	22.97729	-0.07526 $[-0.10234, -0.04914]$
CLIP	29.13552	29.04160	-0.09392 $[-0.19651, -0.00368]$
MUSIQ, full image	45.22784	45.01924	-0.20860 $[-0.70887, +0.31512]$
MUSIQ, 1024 crops	58.32778	57.19612	-1.13166 $[-1.38754, -0.90219]$
CLIP-IQA, full image	0.60299	0.60593	$+0.00294$ $[-0.00018, +0.00609]$
CLIP-IQA, 1024 crops	0.58597	0.58846	$+0.00250$ $[-0.00070, +0.00563]$
Spectrum log RMSE	1.06838	1.06596	-0.00242 $[-0.00905, +0.00433]$
Spectrum slope error	0.19995	0.19741	-0.00254 $[-0.00678, +0.00192]$
Tiling at 1024	0.06836	0.06844	$+0.00008$ $[-0.00072, +0.00081]$

The selected composition has no demonstrated gain on ImageReward or full-resolution MUSIQ. PickScore, crop MUSIQ, and CLIP have negative nominal confidence intervals. The remaining intervals cross zero. This second campaign therefore does not repair the original controller's failure.

The earlier time-matched controller on SEGA's base also regresses on the same 187 seed-0 prompts. Full-resolution MUSIQ changes by -3.03602, with interval $[-4.28169, -1.81488]$. Crop MUSIQ changes by -2.19266, with interval $[-2.81241, -1.56666]$. Its ImageReward difference is unresolved at $+0.01782$ $[-0.02271, 0.05913]$. A temperature-only control on SEGA's base reduces full-resolution MUSIQ by 2.08550 points, with a paired interval $[-3.24526, -0.96407]$. These comparisons constrain the tested designs; they do not show that all spectral controllers must fail.

5 Limits and continuation

Novelty and current comparison scope. A successful continuation requires a distinct mechanism with competitive quality and cost against recent resolution-extrapolation methods. Adaptive spectral gains are already introduced by SEGA; temporal positional schedules by DyPE [5]; entropy-based concentration by UltraImage [17]; and text anchoring with dynamic temperature by TIDE [9]. HRDiT [16] combines spatial position alignment and head-adaptive pruning. More recently, AReA [11] changes attention from coarse cross-scale to progressively local receptive fields, while Spectral Correction Guidance [6] corrects sampling toward a reference spectrum. CSFlow [2] connects contrast sensitivity to flow-time weighting. In language models, DoPE [15] uses rotary-band Gram spectra and truncated

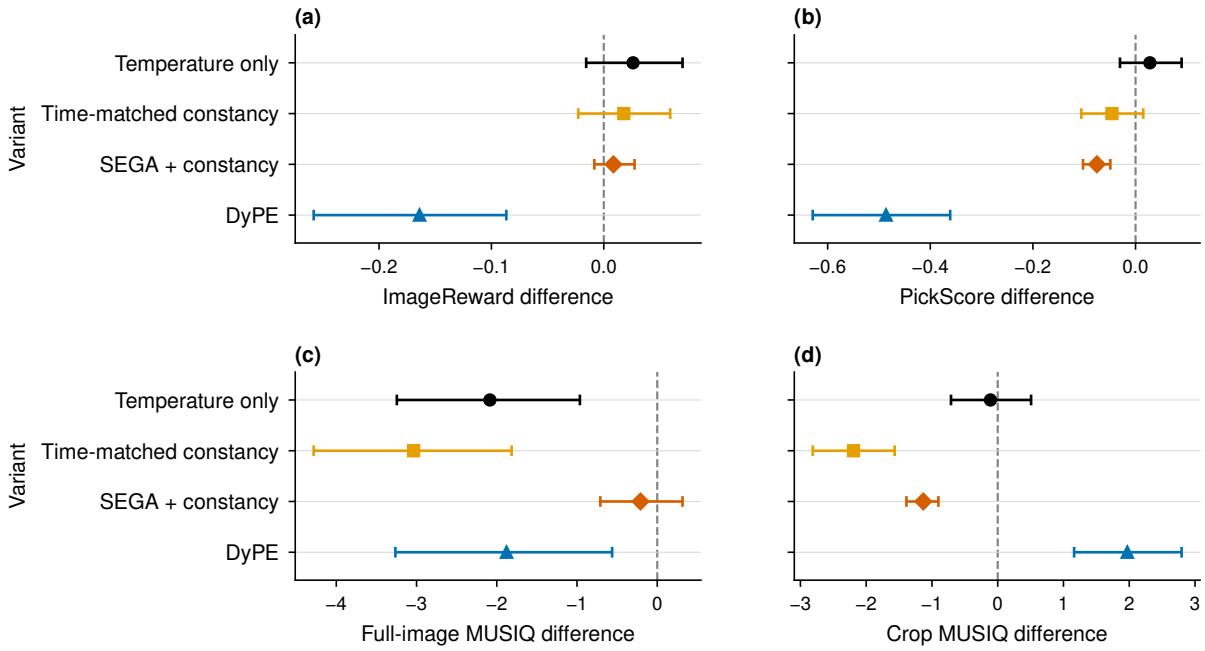


Figure 1: Neither constancy variant demonstrates a joint preference and quality gain over released SEGA. Points show paired differences; whiskers are nominal 95% prompt-bootstrap intervals from 2,000 draws over 187 prompts, single seed. Zero is the released SEGA baseline; higher is better in all panels. DyPE's crop quality gain accompanies a preference loss. The temperature-only and time-matched constancy variants use SEGA's RoPE base and reference temperature.

matrix entropy to select positional interventions. This is a prior-art boundary for covariance-based band suppression, not evidence of its efficacy in DiTs. Thus spectral feedback, perceptual motivation, normalization, or a timing change alone cannot establish the proposed contribution. The current RMS and timing controls are diagnostics. The completed SEGA comparisons do not establish competitiveness against this expanded set, and a development pass alone does not justify promoting a control to a new method. No near-SOTA or SOTA claim follows from the present evidence.

Local response diagnostic. The earlier octave interventions compare complete sampling trajectories, so they do not identify a local map from attention gain to spectral power. A new diagnostic replays a selected transformer state with identical latent, conditioning, and noise level. It perturbs height and width gains separately while preserving the affected axis's gain-vector RMS before attention-dtype quantization, alongside a spatial scalar control. This does not preserve query/key activation energy. Symmetric finite differences estimate a directional response of the clean-estimate spectrum along this normalized path, including BF16 quantization; smaller perturbations check prediction accuracy. Only the baseline output advances the sampler. CPU tests, including a small randomly initialized FLUX transformer, check exact trajectory and state restoration. Both pretrained 2048-resolution smoke tests reproduced the baseline latent endpoint bitwise and passed the same-state no-op replay, but failed the required exact gain-history comparison. The 4K stage therefore did not start, and no response result is accepted under the frozen protocol. A separate four-run repeatability diagnostic reproduced history drift between ordinary baselines. Across both prompts, BF16 gains, recorded intermediate states and velocities, and endpoints matched; the largest observed FP32 gain discrepancy was 1.19×10^{-7}. Repeating allocation on identical projected features also drifted under default arithmetic, while eight repeats with deterministic allocation matched for each prompt. This localizes a numerical repeatability issue to feature-to-gain arithmetic without identifying a unique kernel. A separately frozen diagnostic enables deterministic arithmetic only during allocation and records this numerical-policy change. Both 2048-resolution smoke tests and both 4K runs now pass the original exact replay, gain-history, and endpoint checks. These two smoke endpoints also match those from the original numerical policy. This is a bounded numerical validation, not evidence of general quality equivalence with the original baseline.

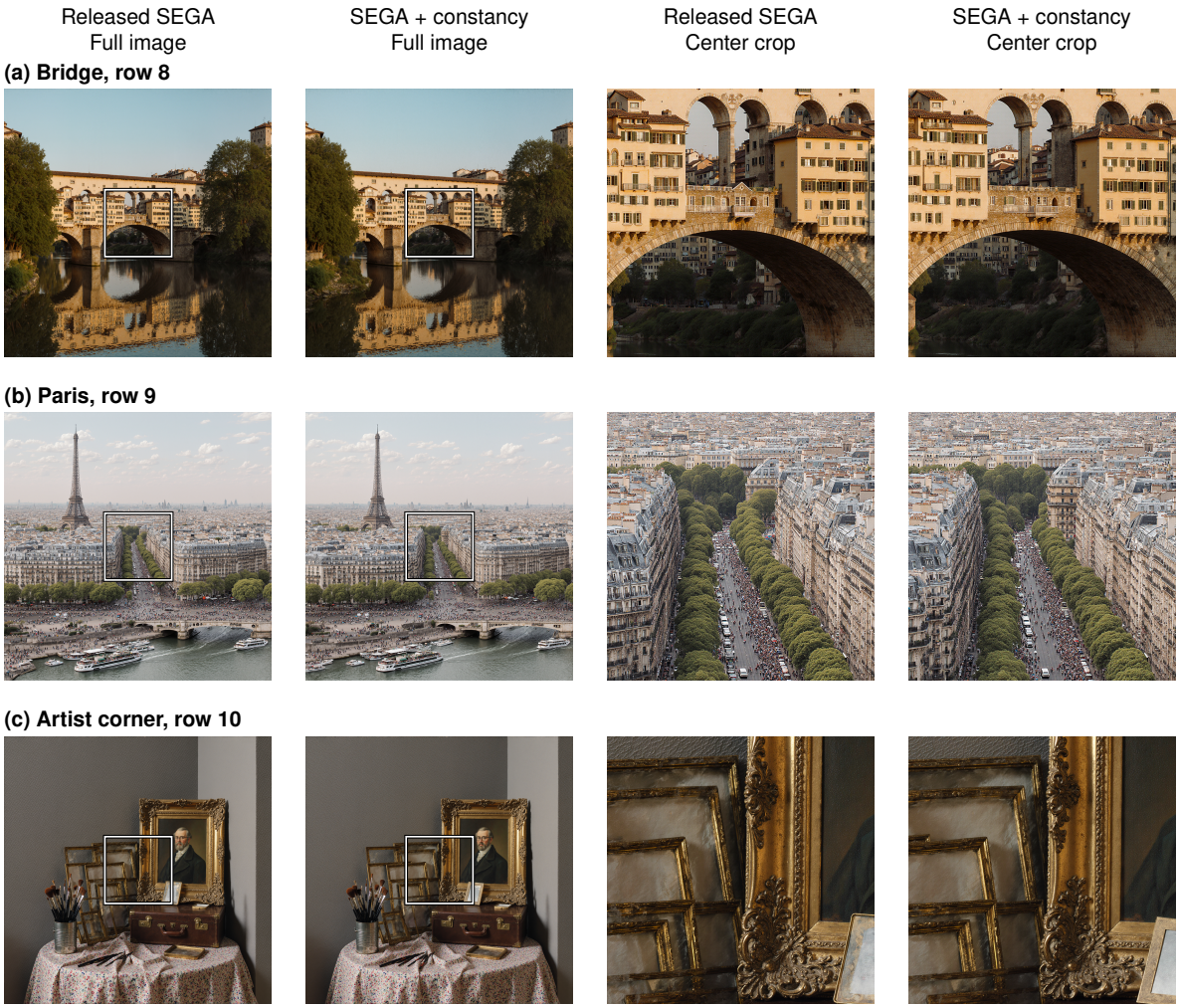


Figure 2: Fixed-order examples from the completed second campaign: the first three held-out captions, rows 8–10, seed 0. Columns compare released SEGA with SEGA plus constancy at $\eta = 1$, followed by their native 1024^2 center crops. Boxes show the crop coordinates in the 4096^2 full outputs. The same pixel coordinates are used in each image; generated content need not align exactly. Full images are reduced for display, and crops receive no enhancement. Rows were fixed before viewing these images and were not selected by scores. Three examples cannot establish preference or quality differences; the paired 187-prompt measurements supply the quantitative evidence.

On two development prompts, we probe steps 6, 14, and 22 of 28 with full perturbation magnitude 0.10 and unfit half magnitude 0.05. For each spatial axis, the selected rotary octave is -11; it lies below the minimum nonzero spatial DFT frequency of the token grid. These measurements therefore do not characterize all rotary bands. Prediction error is the norm of the combined positive and negative half-step residuals divided by the corresponding observed response norm; a value of one matches a zero-response predictor. Across the 12 axis-octave cases, the median error is 0.937 (range 0.709–1.354), with nine cases below one. Across six spatial-scalar cases, the median is 0.426 (range 0.050–3.039). The late scalar errors are 0.104 and 0.050, but the early failure on one prompt prevents a uniform predictability claim. Raw response magnitudes are not directly comparable because the actuator norms differ. These results do not support inverting the current octave response map. An unprespecified secondary analysis uses the actual BF16 log-gain changes as control coordinates. It fits only full-magnitude observations and predicts the untouched half-magnitude responses. Axial errors remain 0.688–1.339 and scalar errors 0.029–3.004. No intervention vanishes entirely through quantization. However, 31–49% of each axial half-step gain-change norm lies outside the full-step secant direction (11–16% for scalar controls). A one-dimensional coordinate correction therefore does not separate quantization from nonlinear response; it does not change the frozen diagnostic's acceptance rules. The diagnostic adds 39 transformer calls per 4K trajectory; recorded baseline and probed sampling times are 201/485 and 185/441 seconds for the two prompts. This is measurement overhead, not an efficient generation method.

This is an instrument for testing the controller hypothesis, not a new validated generation method. SCG acts directly on latent Fourier magnitudes, whereas this diagnostic measures the response to internal attention gains [6, 12]; this difference alone does not establish novelty or improved quality. Jacobian-based analysis and correction are established: Graikos et al. study denoiser-input Jacobian spectra and train with response regularization, while JAPS uses denoiser Jacobian information for inverse-problem guidance [3, 4]. Our measured object is instead the response of Fourier-power statistics to attention gains at a fixed state. Attention Frequency Modulation already edits spatial frequency bands of cross-attention logits and uses entropy-dependent gating [10]. Thus spectral attention control itself is also established. The proposed response map remains a hypothesis to test, not sufficient evidence of a novel controller. Local prediction must also transfer to the final image at acceptable sampling cost. The failed two-prompt smoke test is retained as an implementation finding, not a result about generation quality or the proposed response model.

Observer invariance diagnostic. A separate source audit identifies a limitation of the released SEGA observer. For the standard text-conditioned FLUX path, with fixed attention gains and no optional adapters, each residual branch reads a LayerNorm-normalized state and the output projection follows a final LayerNorm. In real arithmetic, $\mathrm{LN}(h_i + a_i\mathbf{1}) = \mathrm{LN}(h_i)$ for a scalar a_i at each token. Residual additions preserve this common-channel shift until the final normalization removes it. Such residual-stream centering is established in transformer analysis [14]. However, SEGA takes the spatial spectrum of the channel mean of the initial projected features. That mean changes by a_i, so the observer can change even when the fixed-gain denoiser function does not. This argument concerns a representation dependence, not proof of poor quality on the released weights. Finite-precision reparameterizations require separate checks. On a randomly initialized miniature FLUX with one joint and one single block, three common-channel reparameterizations preserve the output within 1.23×10^{-15} maximum absolute error while changing or zeroing the observer's spatial energy. A channel-dependent negative control preserves the channel mean but changes the output by 2.025 maximum absolute error. Parameters and outputs are FP64; the installed rotary implementation includes FP32 casts. Seven CPU checks pass. This is an architecture-level numerical counterexample, with no pretrained, BF16, trajectory, or image-quality result. An independent miniature test uses the released SEGA observer and the installed gain processor on a 4×128 token grid. A spatially varying common-channel intervention changes gains by 0.1175 and the adaptive output by 0.00717 maximum absolute difference. Keeping original gains reduces the output difference to 1.34×10^{-15}; applying the changed gains to original features reproduces the shifted adaptive output to the same tolerance. Seven additional checks pass. This isolates gain mediation in the miniature; the intervention is on activations at one input, not a universal parameter reparameterization.

A separately frozen pretrained BF16 audit evaluates two existing development prompts at 4096^2. Each input is an artificial mixture of a validated sampling endpoint and fixed CPU-generated Gaussian noise at $\sigma = 0.5$, rather than a recorded reverse-trajectory state. The intervention replaces the projected channel-mean map with a fixed spatial harmonic while matching its original mean and population standard deviation before BF16 rounding. The original and shifted features are crossed with their respective SEGA gains; one additional original-state repeat gives five forwards per prompt, ten in total. Saved original/repeat outputs match byte for byte, and provenance, gain crossings, saved tensors, and output contrasts pass independent CPU validation.

Let F_{ab} denote the velocity with original ($a = 0$) or shifted ($a = 1$) features and original ($b = 0$) or shifted-observer ($b = 1$) gains. The gain-only effect $\|F_{01} - F_{00}\|_2$ is 109.986 and 66.181 for the two prompts, compared with fixed-gain residuals $\|F_{10} - F_{00}\|_2$ of 16.505 and 16.448. Their ratios are 6.664 and 4.024. The adaptive changes $F_{11} - F_{00}$ align with the gain-only changes with cosines 0.988 and 0.963. However, $\|F_{11} - F_{01}\|_2/\|F_{01} - F_{00}\|_2$ remains 0.156 and 0.271, and the interaction residual has relative magnitude 0.215 and 0.366 against the gain-only effect. Thus the pretrained intervention has a substantial gain-mediated component, with non-negligible finite-precision or architectural residuals; it does not reproduce exact mathematical invariance. Full projected features and conditioning were not saved, so their checks remain runtime receipts. These two synthetic-state measurements do not establish a defect on natural trajectories, an image-quality effect, or a new method.

Changing the observer to channel-centered spectral energy would remove this particular dependence, but would not itself establish novelty or quality. AReA already rejects channel means because opposite-sign feature changes can cancel, and uses channel standard deviation before its wavelet detail measurement [11]. Its statistic is also invariant to a common-channel shift in real arithmetic. The SEGA-specific audit establishes sensitivity under the declared intervention, not a new-method claim.

Spatial-complement attention diagnostic. A separate passive capture tests whether spatial complement statistics justify further method development. PISA already formulates a first-order correction using the mean within-block value–key cross-moment over unselected blocks, then replaces it with an all-block mean for efficiency [7]. Computing the conditional mean is therefore a refinement of a known expression, not a new attention approximation.

We reuse the two artificial inputs above with their original gains. One unmodified forward and one capture forward per prompt give four forwards in total; their final outputs match byte for byte. We retain the actual BF16 post-gain queries, keys, values and attention outputs at calls 0, 18 and 56, heads 0 and 12, and 16 fixed image queries. Independent FP64 evaluation uses these saved tensors and the same dense softmax target. Image tokens form 16×16 cells; the exact set contains cells within Chebyshev distance two of the query cell and all text tokens. All remaining cells share a centroid mass approximation. We compare zeroth-order values, global or complementary mean cross-moments, and individually weighted cell cross-moments.

Across all 192 fixed cases, median relative output errors are 0.3133 for exact sparse attention, 0.0400 for zeroth-order completion, 0.0203 for the global first-order mean, 0.0192 for the complementary mean, and 0.0116 for the weighted cell correction. The corresponding maxima are 0.9297, 0.6063, 0.5930, 0.5940, and 0.5349. These are descriptive summaries of this stencil, not a representative benchmark or a reproduction of PISA's routing and fused kernels. The complementary mean improves 138 cases and worsens 54. Its absolute error change exceeds the separately measured BF16-versus-FP64 attention gap in only 33 cases: 26 improvements and seven regressions. Errors remain large in the sampled final joint block, including with individually weighted moments. Independent blockwise calculations agree with all reported outputs within 3.91×10^{-14}. The small, mixed conditional-mean benefit does not justify a GPU implementation or a generation campaign for this branch. No image-quality, resolution-extrapolation, speed, or novelty claim follows.

Native-resolution capability screen. We generated one 1024^2 comparator for each of the eight existing development captions, using the same model revisions, prompt-derived seed, 28-step sigma schedule and guidance as the archived 4096^2 SEGA images. At native resolution, the deployed SEGA implementation has unit rotary gains and its base positional mapping. All eight images, sidecars, saved initial transformer inputs and archived 4K references pass independent provenance checks. Different noise shapes are uncoupled, and the deployed positional mapping and gains change with resolution. These pairs therefore support an observational capability screen, not an isolated causal effect of resolution.

Before seeing the native outputs, we fixed one primary caption predicate and one secondary observation per caption, with two separately reported shop-text checks. Two model reviewers, GPT-6 Astra and GPT-6.1 Sol, independently inspected all 16 images in a randomized, condition-label-hidden presentation at a common 1024-pixel display size. Each saved its complete common-view record before opening original-resolution images. Both then inspected five originals covering every ambiguous or violated predicate and both text checks; clear observations for the other eleven views were carried forward explicitly. Reviewer A also inspected four recorded crops. This is model-based annotation, not a human study, and visible detail can reveal resolution despite hidden labels.

Row	Caption check	Primary native/4K	Secondary native/4K
0	Distant larger boats; small-boat colors	S/S	S/S
1	Sunset below stars; Moai	S/S	S/S
2	Offering grapes; kneeling by bear	S/S	V/S
3	Yellow board supports surfer; coral below	S/S	S/S
4	Bodies near rings and sun; foreground planet	V/U	S/S
5	Black top and green striped skirt; man and boy	S/S	S/S
6	Holding hands; shared teal couch	S/S	S/S
7	One supporting leg; yellow dress	S/S	S/S

S denotes satisfied, V violated, and U unjudgeable. Both reviewers gave these same labels at common scale, and detail inspection changed none. The full primary native-by-4K transition matrix has seven S/S pairs and one V/U pair; all other seven cells are zero. In the celestial scene, the native image lacks the required smaller body near the sun, whereas the possible sun's identity is ambiguous in the 4K image. The secondary kneeling check improves from V to S: the native woman squats. Auxiliary text is separate: both reviewers label *Kodak* V/V and *Paradis* S/V, including after detail inspection. No annotation disagreement was resolved or suppressed.

This screen provides no primary native-success/4K-failure example to motivate the proposed semantic-loss intervention. We close that particular rationale without a seed search or text-repair pivot. Eight selected development captions, one draw per condition and two model readings cannot estimate a population failure rate or exclude other resolution failures. No new method or quality improvement is established, and the fresh confirmation set remains unused.

The second-campaign candidate has one held-out seed and one evaluated backbone at one resolution. No second-seed replication of this candidate, new backbone, human preference study, or broader resolution claim is supported. Aggregate FID/KID scores from the raw 188-image evaluation include development row 7 and are excluded from the held-out conclusion. No new throughput comparison is claimed in this report.

The earlier extended theory and literature sections remain in the source tree but are excluded because they describe superseded settings and unresolved claims. Section 2 retains only qualified algebraic observations. The third diagnostic below tests gain magnitude, frequency allocation, and controller timing under a frozen development-only protocol. The already inspected held-out set must not become a tuning set presented as fresh evaluation.

Third development diagnostic. The three prespecified controls and a fresh released-SEGA baseline completed on eight development prompts, seed 0. These are exploratory selection scores, not held-out efficacy evidence.

Table 3: Third development diagnostic. Differences are against the fresh same-hardware SEGA baseline.

Variant	ΔIR	ΔMUSIQ	ΔCLIP	Δcrop MUSIQ	Gate
RMS-matched uniform	-0.0174	-1.5667	+0.5868	-0.5177	Fail
RMS-normalized allocation	+0.0363	+0.3889	-0.0655	-0.2150	Pass
Lag-aligned constancy	+0.0193	-0.2233	+0.0092	-0.1530	Fail

PickScore differences against the same baseline are RMS-matched uniform: -0.0832; RMS-normalized allocation: -0.0666; Lag-aligned constancy: -0.0546. RMS-normalized allocation passed the development gate. These controls remain diagnostics, not an established novel method. A separate set of 64 confirmation captions was frozen before these outcomes; its use requires a distinct method and a recent-baseline protocol. No confirmation result or efficacy claim follows from this table.

Conditional confirmation protocol. Before the third development outcome, a separate set of 64 captions was frozen from Aesthetic-4K's 2048-source split after excluding normalized matches to all prior evaluation and calibration captions. Fixed hash ordering selected the captions without inspecting generation or scores. A candidate that satisfies both the method-novelty review and a separately frozen development protocol would use its unchanged settings at 4096^2 with a fresh SEGA baseline. The captions are fresh to this project's evaluation, with no model-training disjointness or semantic-near-duplicate guarantee. They do not supply matched 4K reference images. The prespecified analysis uses 10,000 paired prompt-bootstrap draws, seed 0, and common resampling indices across metrics and candidates. Nominal 95% intervals accompany all metrics. For the two primary metrics, ImageReward and full-image MUSIQ,

Bonferroni-adjusted percentile intervals have coverage $1-0.05/(2K)$ for K selected candidates. Joint primary support requires both adjusted lower bounds to exceed zero. These are approximate bootstrap intervals, not exact finite-sample guarantees. Secondary losses remain part of the report; confirmation cannot select further controller settings. No confirmation result is available in this snapshot.

Evidence provenance. The completed campaign is recorded at implementation commit `a1ede1a` on October 16, 2026. Its archived paired table, per-prompt scores, and analysis protocol are in `.agent/notes/main2/`. The current held-out table is generated from those CSVs after checking its means and paired intervals against the per-prompt records. This check reuses saved scorer outputs; it is not a fresh image-scoring reproduction. The previous session records 194 passing CPU tests. The third diagnostic adds new inference and evaluation; its development-only outcome is reported separately above.

6 Prospective image-to-text phase intervention

We define a candidate that changes the conditional distribution over text keys while preserving the deployed image-key probabilities and total text probability at each image-query row, in real arithmetic. The hypothesis is that cross-modal routing may benefit from a different rotary geometry than image-to-image attention. The native capability screen supplies no primary semantic failure that this candidate is known to repair.

For an image-token grid of height H and width W, use the fixed reference map

$$\Phi_{H,W}(p_0, p_h, p_w) = \left(p_0, \frac{63p_h}{H-1}, \frac{63p_w}{W-1}\right). \tag{8}$$

Thus a 256×256 grid maps its spatial endpoints to 0 and 63; the map is the identity at 64×64. The reference denotes our nominal 1024^2 setting, not a verified boundary of the model's training coordinates. At each block we rotate the same normalized pre-RoPE image query using base frequencies with $\theta = 10000$ and axis dimensions $(16, 56, 56)$ at $\Phi_{H,W}(p)$. We retain the actual pair-equal amplitude gains, text keys and all values. Deployed SEGA geometry [12] still determines image-key logits. The intervention covers all 24 heads, all 57 joint and single blocks, all 28 denoising steps and all 512 text slots, including padding. Runtime text IDs must be zero. Text-query outputs remain unchanged within that block; later text activations can change through earlier modified image states.

For one image query, let a_j and b_k be the deployed, scaled logits for text keys T and image keys I. Define $Z_T = \sum_{j \in T} e^{a_j}$, $Z_I = \sum_{k \in I} e^{b_k}$, and $m = Z_T/(Z_T + Z_I)$. If a'_j are the transported-phase text logits, set

$$\widetilde{a}_j = a'_j + \mathrm{LSE}(a) - \mathrm{LSE}(a'), \qquad b'_k = b_k. \tag{9}$$

The text partition remains Z_T, so every image-key probability and total text mass m are unchanged. With $u_T = \mathrm{softmax}(a)V_T$ and $u'_T = \mathrm{softmax}(a')V_T$, the equivalent output correction is

$$y' = y + m(u'_T - u_T). \tag{10}$$

These are elementary softmax identities. They impose a local constraint and do not establish an improvement in the sampling trajectory.

The unnormalized control uses the same phase map but permits text mass to change. It computes the original image-only conditional output u_I and log-partition ℓ_I, and the new text-only output u'_T and log-partition ℓ'_T. For $\ell = \mathrm{logaddexp}(\ell_I, \ell'_T)$,

$$y_{\mathrm{unc}} = e^{\ell_I - \ell} u_I + e^{\ell'_T - \ell} u'_T. \tag{11}$$

The original joint call supplies unchanged text-query outputs. This stable mixture avoids recovering a small image contribution by subtracting rounded text output from a BF16 joint output. Its extra image-only pass is a declared cost. At 65,536 image tokens and 512 text slots, the fixed-mass arm adds approximately 1.538% of baseline joint-attention QK/AV arithmetic through two text branches; the unnormalized control adds approximately 99.225% through image-only and new-text branches. These counts exclude rotations, buffers and kernel overhead, and are not latency predictions.

All three arms use a pinned private cuDNN output-and-log-sum-exp backend with no fallback. The protocol requires tiny BF16 kernel comparisons against FP64 oracles and a production-shaped text-branch support probe before

model loading. Every full-resolution call must satisfy the backend's output/LSE contract. A four-step 1024^2 smoke compares the deployed attention path with the new baseline, accepting endpoint relative L_2 error at most 10^{-3} and recording cosine diagnostically. Both native phase arms must reproduce the matched baseline endpoint bytes. Native execution shares the baseline path; separate non-bypassed CPU oracles check the phase and partition formulas. The fixed-mass correction accumulates in FP32 before its BF16 output cast. Real-arithmetic conservation therefore remains distinct from floating-point equivalence.

Positional interpolation [1], cross-modal geometry separation in Circle-RoPE,[1] query-dependent phase scales in Phase-Aligned RoPE,[2] and token-group rotary edits in Untwisting RoPE[3] are established ingredients. TACA already changes relative text-token odds through modality-specific temperature,[4] and TIDE studies text/image imbalance [9]. The proposed composition isolates phase changes while holding text mass fixed; that constraint also prevents it from correcting pure text-mass dilution. Neither novelty nor a quality advantage has been established.

The frozen development pilot specifies exactly six 4096^2 images: two existing captions, seed 0, 28 steps and guidance 4.5 under the matched baseline, fixed-mass and unnormalized arms. All use the same Krea backbone in BF16 and must share exact actual initial transformer inputs within each prompt and resolution. Both captions already satisfy their primary predicates, so this pilot tests regression, detail and cost with little primary-semantic headroom. Its scope excludes a seed search and fresh confirmation data; any readout must retain all six images and report the unnormalized control's extra cost.

References

[1] Shouyuan Chen, Sherman Wong, Liangjian Chen, and Yuandong Tian. Extending context window of large language models via positional interpolation, 2023. arXiv:2306.15595.

[2] Malgorzata Galinska, Bart Pogodzinski, and Jan Eric Lenssen. CSFlow: Aligning flow matching with human contrast sensitivity, 2026. URL https://arxiv.org/abs/2606.08833.

[3] Alexandros Graikos, Nebojsa Jojic, and Dimitris Samaras. On the spectral properties of generative denoiser jacobians. *arXiv preprint arXiv:2609.36210*, 2026. URL https://arxiv.org/abs/2609.36210.

[4] Liav Hen, Tom Tirer, Raja Giryes, and Shady Abu-Hussein. Jacobian-aware posterior sampling for inverse problems. *arXiv preprint arXiv:2511.18471*, 2026. URL https://arxiv.org/abs/2511.18471v3. Version 3, June 21, 2026.

[5] Noam Issachar, Guy Yariv, Sagie Benaim, Yossi Adi, Dani Lischinski, and Raanan Fattal. DyPE: Dynamic position extrapolation for ultra high resolution diffusion, 2025. arXiv:2510.20766.

[6] Gihoon Kim and Taesup Kim. Correcting guided diffusion trajectories with spectral alignment, 2026. URL https://arxiv.org/abs/2610.02753.

[7] Haopeng Li, Shitong Shao, Wenliang Zhong, Zikai Zhou, Lichen Bai, Hui Xiong, and Zeke Xie. PISA: Piecewise sparse attention is wiser for efficient diffusion transformers. *arXiv preprint arXiv:2602.01077*, 2026. URL https://arxiv.org/abs/2602.01077v2. Version 2.

[8] Xingchao Liu, Chengyue Gong, and Qiang Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2209.03003.

[9] Yihua Liu, Fanjiang Ye, Bowen Lin, Rongyu Fang, and Chengming Zhang. TIDE: Text-informed dynamic extrapolation with step-aware temperature control for diffusion transformers, 2026. URL https://arxiv.org/abs/2603.08928.

[10] Seunghun Oh and Unsang Park. Attention frequency modulation: Training-free spectral modulation of diffusion cross-attention. *arXiv preprint arXiv:2603.28114*, 2026. URL https://arxiv.org/abs/2603.28114.

[1] Wang et al., *Circle-RoPE: Cone-like Decoupled Rotary Positional Embedding for Vision-Language Models*, arXiv:2505.16416v3.
[2] Wu et al., *Phase-Aligned RoPE for Mixed-Resolution Diffusion Transformer*, arXiv:2511.19778v2.
[3] Mikaeili et al., *Untwisting RoPE: Frequency Control for Shared Attention in DiTs*, arXiv:2602.05013v1.
[4] Lv et al., *Rethinking Cross-Modal Interaction in Multimodal Diffusion Transformers*, arXiv:2506.07986v3.

[11] Javad Rajabi, Koorosh Roohi, David B. Lindell, and Babak Taati. AReA: Adaptive receptive-field attention for efficient high-resolution diffusion transformers, 2026. URL https://arxiv.org/abs/2610.17227.

[12] Javad Rajabi, Kimia Shaban, Koorosh Roohi, David B. Lindell, and Babak Taati. SEGA: Spectral-energy guided attention for resolution extrapolation in diffusion transformers, 2026. URL https://arxiv.org/abs/2605.22668. arXiv:2605.22668.

[13] Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. RoFormer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568, 2024. arXiv:2104.09864.

[14] TransformerLens contributors. Weight processing: Centering residual-stream writing weights. TransformerLens documentation, n.d. URL https://transformerlensorg.github.io/TransformerLens/generated/code/transformer_lens.weight_processing.html. Accessed October 17, 2026.

[15] Jing Xiong, Liyang Fan, Hui Shen, Zunhai Su, Min Yang, Lingpeng Kong, and Ngai Wong. DoPE: Denoising rotary position embedding, 2025. URL https://arxiv.org/abs/2511.09146v1.

[16] Yu Xue, Haoxuan Qu, Zhuoling Li, Hongbin Xu, Jianxiong Yin, Simon See, Hossein Rahmani, and Jun Liu. HRDiT: Training-free high-resolution image generation with off-the-shelf diffusion transformer models, 2026. URL https://arxiv.org/abs/2608.07003.

[17] Min Zhao, Bokai Yan, Xue Yang, Hongzhou Zhu, Jintao Zhang, Shilong Liu, Chongxuan Li, and Jun Zhu. UltraImage: Rethinking resolution extrapolation in image diffusion transformers, 2025. URL https://arxiv.org/abs/2512.04504.